UNIVERSITÀ DEGLI STUDI DI TRIESTE

Dipartimento di Ingegneria e Architettura

Simulating Mechazilla's catching system

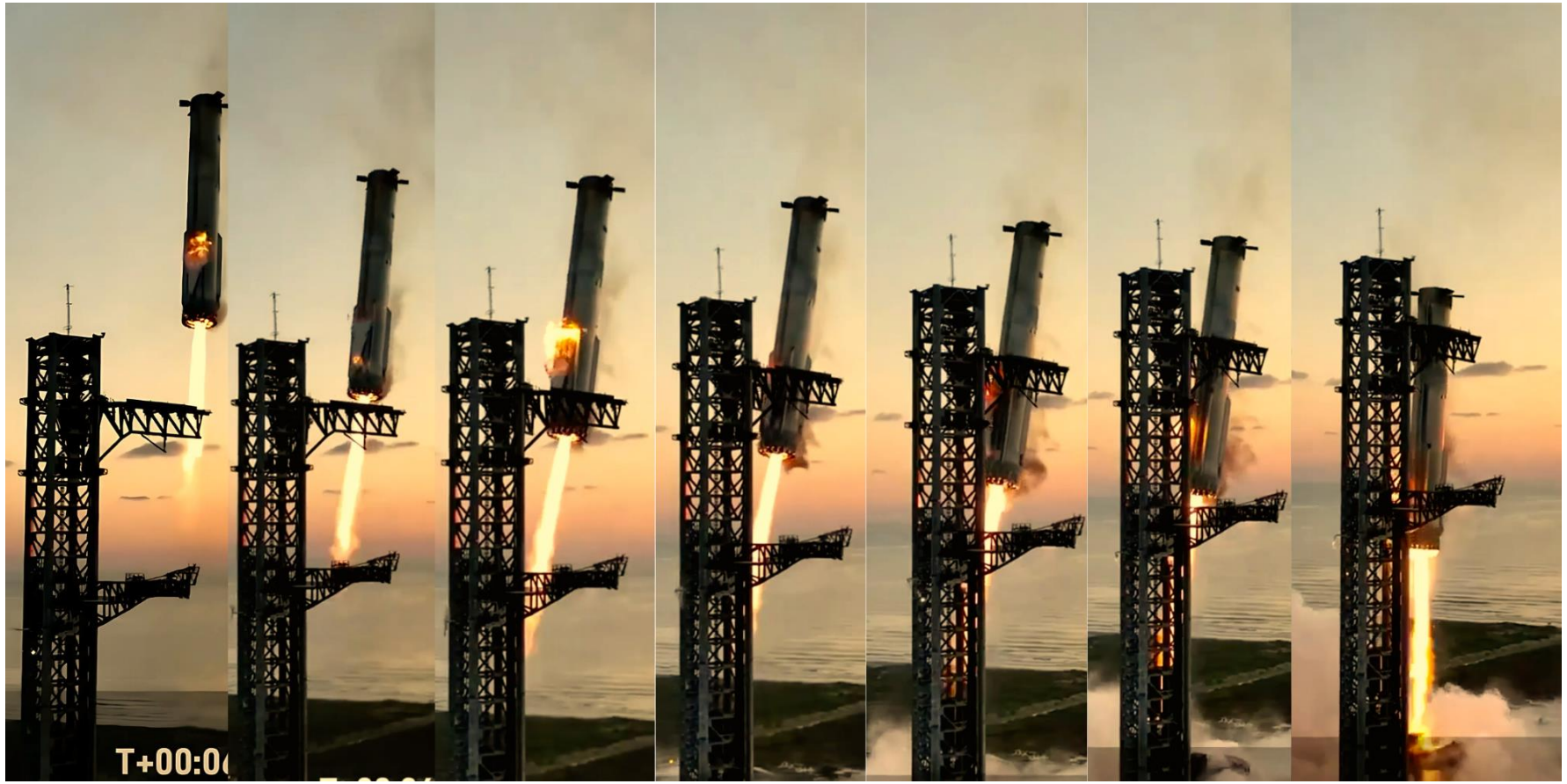
Learning-based Control project

Student:
Michelangelo Pozza

ID:
IN2300012

Lecturer:
Prof. Erica Salvato

The idea behind the project



The idea behind the project

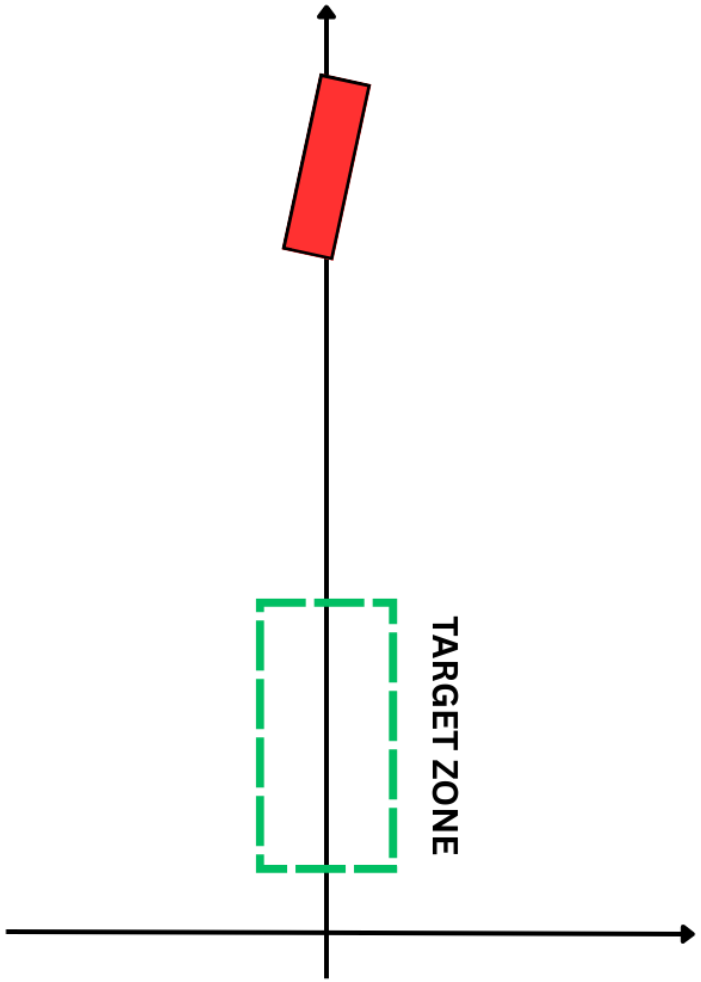
- A spacecraft is returning from space
- Avoiding a crash, and a loss of milions, in the ocean is a priority
- SpaceX came in with a big tower equipped with a complex mechanic claw system to catch a 4000 tons booster
- The engineering complexity of such a system is far too complex from our course objective, and Mars colonization too...

What if the spacecraft try alone to learn how to land using RL?

The task

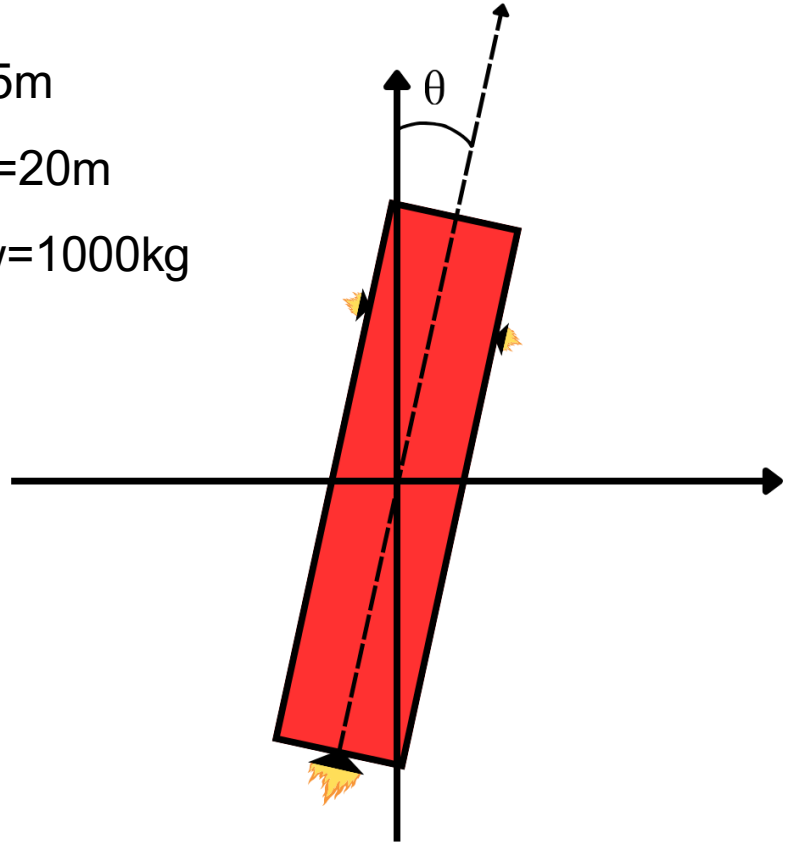
Our challenge is to stabilize the rocket within the 'catch zone' with a minimal inclination angle and the lowest possible velocity to ensure the structure remains undamaged

NOTE: The spacecraft has a maximum of 20 seconds to land (free fall takes approximately 10s)



The system

- Base b=5m
- Height h=20m
- Weight w=1000kg



- The spacecraft is approximated as a rectangle for simplicity
- We have three booster:
  - One main booster for altitude control
  - Two lateral thrusters for angle adjustment
- This simplified version of the problem consists in controlling only the spacecraft angle and altitude

State space description

$$\mathbf{x} = \begin{bmatrix} p_y \\ \theta \\ \dot{p}_y \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

$$\mathcal{X} = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \in \mathbb{R}^4 \right\}$$

Physical constraints

$p_y \in [\frac{h}{2}m, 200m]$ $\quad$ $\theta \in [-12°, 12°]$

$\dot{p}_y \in [-50m/s, 50m/s]$ $\quad$ $\dot{\theta} \in [-1rad/s, 1rad/s]$

Target constraints

$p_y \in [30m, 40m]$ $\quad$ $\theta \in [-5°, 5°]$

$\dot{p}_y \in [-5m/s, 5m/s]$ $\quad$ $\dot{\theta} \in [-0.1rad/s, 0.1rad/s]$

NOTE: the position is relative to the rectangle's centroid!

Action space description

$$u = \begin{bmatrix} F_c \\ F_l \\ F_r \end{bmatrix}$$

$$\mathcal{U} = \left\{ \begin{bmatrix} F_c \\ F_l \\ F_r \end{bmatrix} \in \mathbb{R}^3 \right\}$$

Physical constraints

$$F_c = \{0, 20kN\}$$
$$F_l = \{0, 5kN\}$$
$$F_r = \{0, 5kN\}$$

Input combination

$$\begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} F_{c_{on}} \\ F_{l_{on}} \\ F_{r_{on}} \end{bmatrix}$$

The lateral boosters are located at y=h/4 relative to the centroid

System dynamics

$$\dot{\mathbf{x}} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} x_3 \\ x_4 \\ \frac{1}{m}\left[(F_l - F_r)\sin(x_2) + F_c\cos(x_2)\right] - g \\ \frac{(F_r - F_l)\frac{h}{2} - d_\theta x_4}{I} \end{bmatrix}$$

$$I = \frac{m(b^2 + h^2)}{12}$$

Moment of inertia of the rectangle (axis perpendicular to the plane)

$$d_\theta = 20000\, N \cdot m \cdot \frac{s}{rad}$$

Angular damping coefficient

Reward

- *Potential based reward* because of local maximum and strong physics
- The potential is simply the *L2 norm* of the difference between the values and their target
- The matrix Qr uses *Bryson's rule* to normalize all measurements
- A *constant step penalty* of -0.1 is included to encourage the agent to stabilize quickly

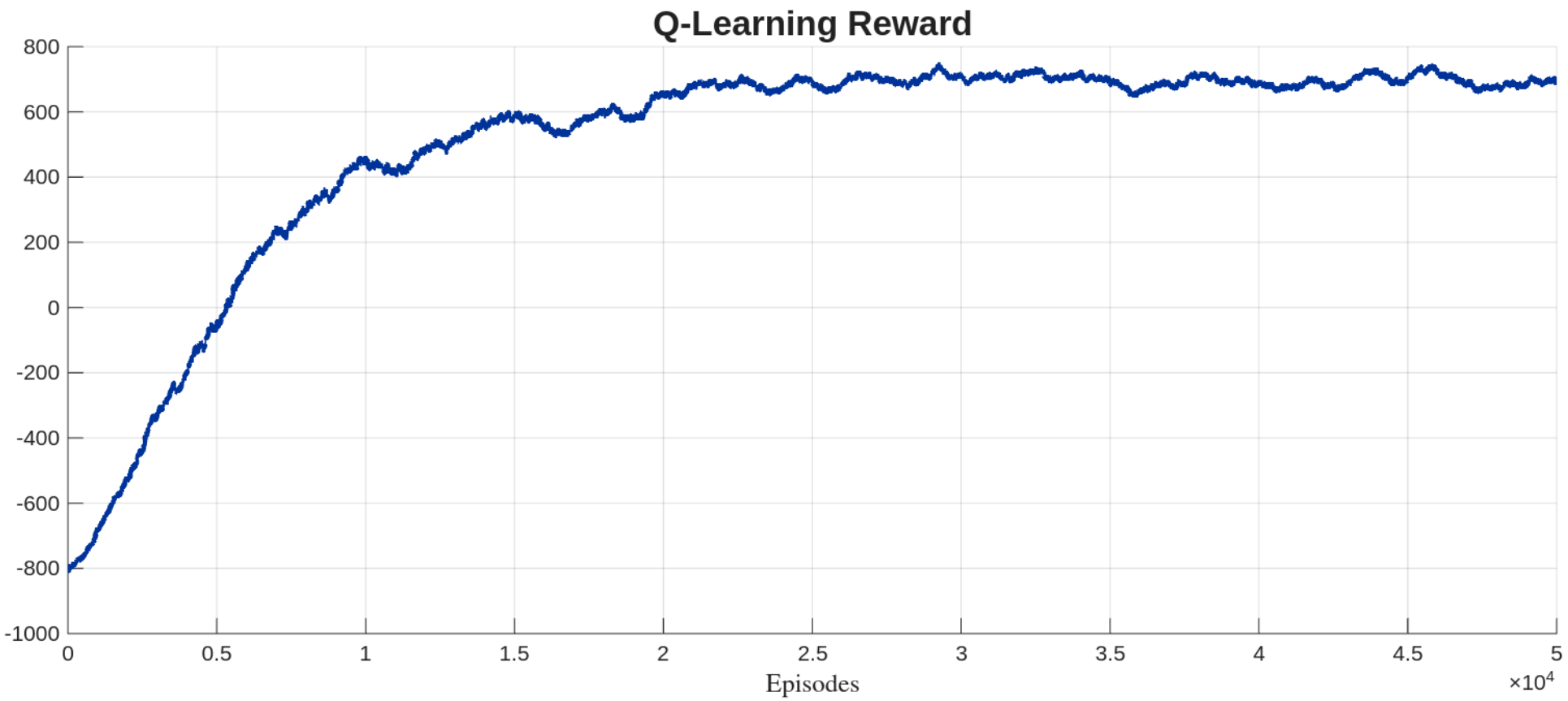
$$r(k+1) = \gamma\phi(x(k+1)) - \phi(x(k)) - 0.1$$

$$\phi(s) = -||x(k) - x_{des}||_2$$

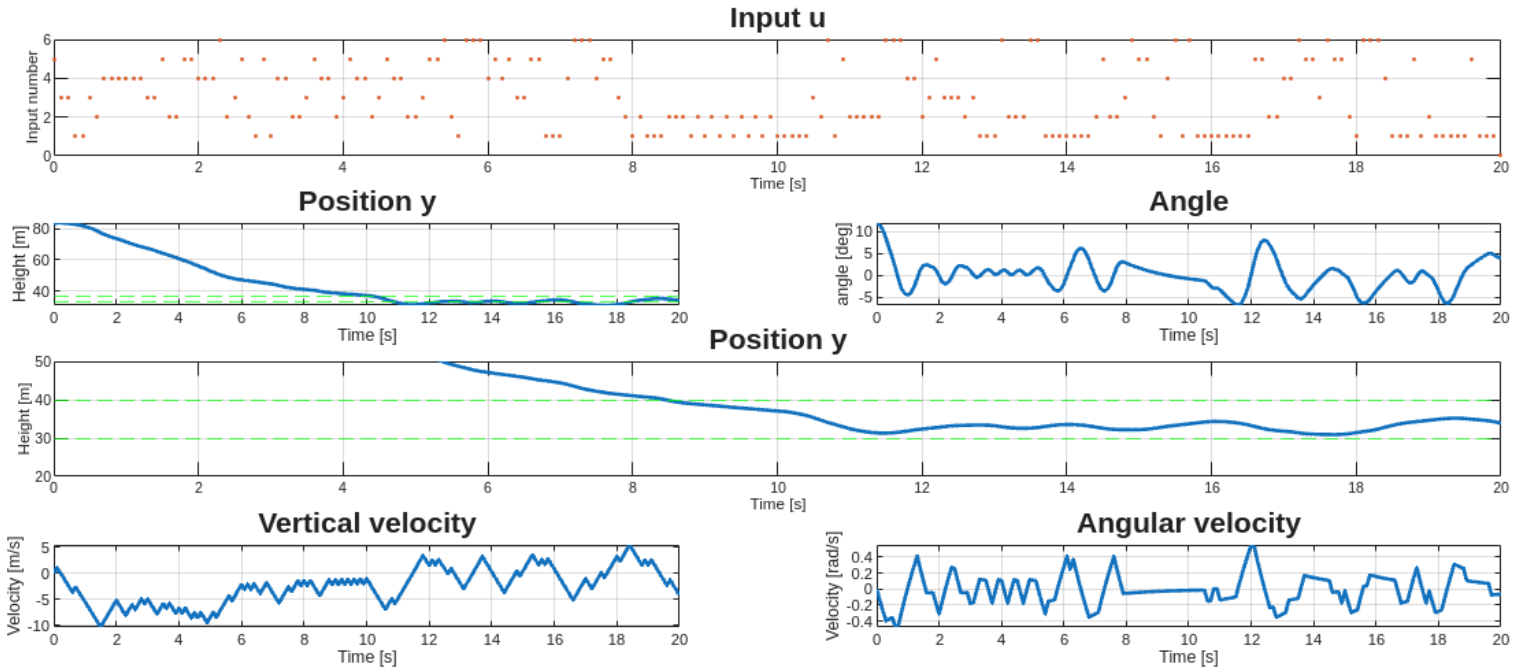
$$= -\sqrt{(\mathbf{x} - \mathbf{x}_{des})^T Q_r (\mathbf{x} - \mathbf{x}_{des})}$$

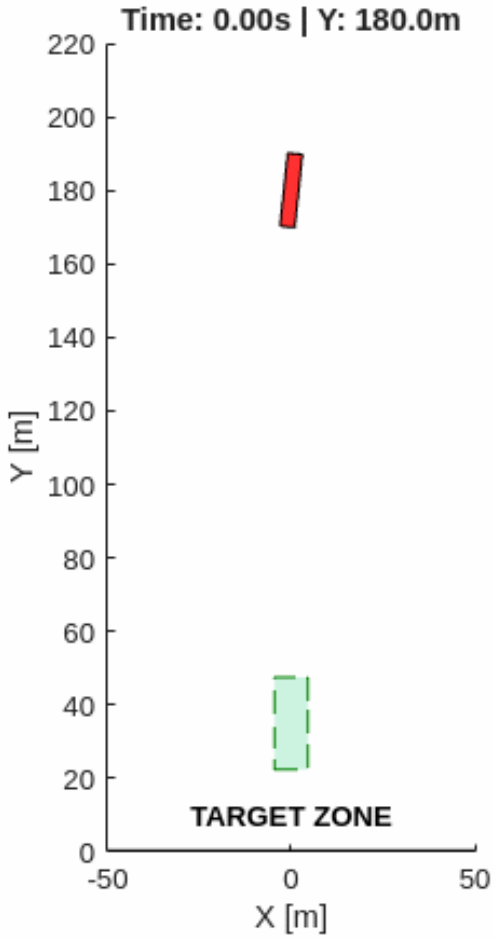
$$Q_r = \begin{bmatrix} \frac{1}{x_{1,max}^2} & 0 & 0 & 0 \\ 0 & \frac{1}{x_{2,max}^2} & 0 & 0 \\ 0 & 0 & \frac{1}{x_{3,max}^2} & 0 \\ 0 & 0 & 0 & \frac{1}{x_{4,max}^2} \end{bmatrix}$$

Q-Learning - training



Q-Learning - testing



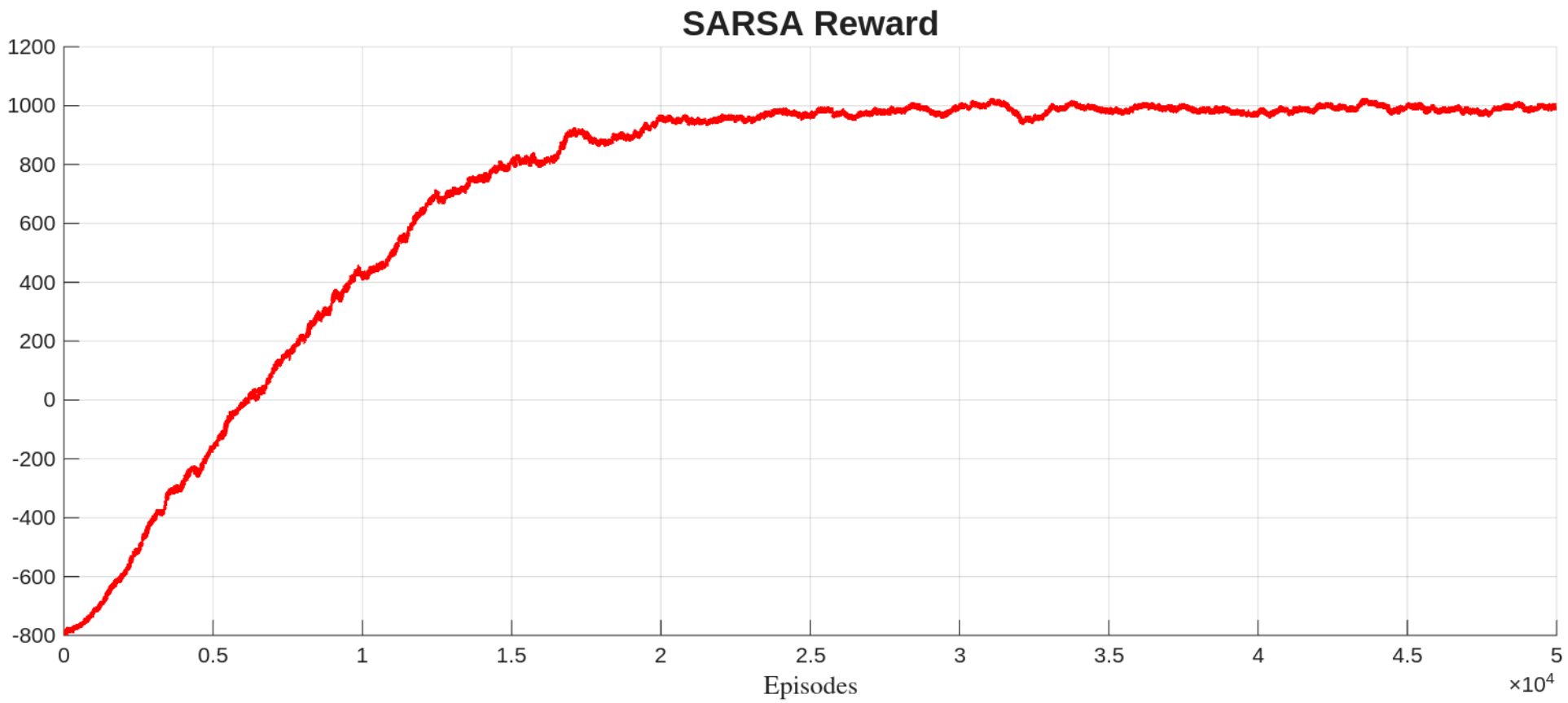


Initial conditions:

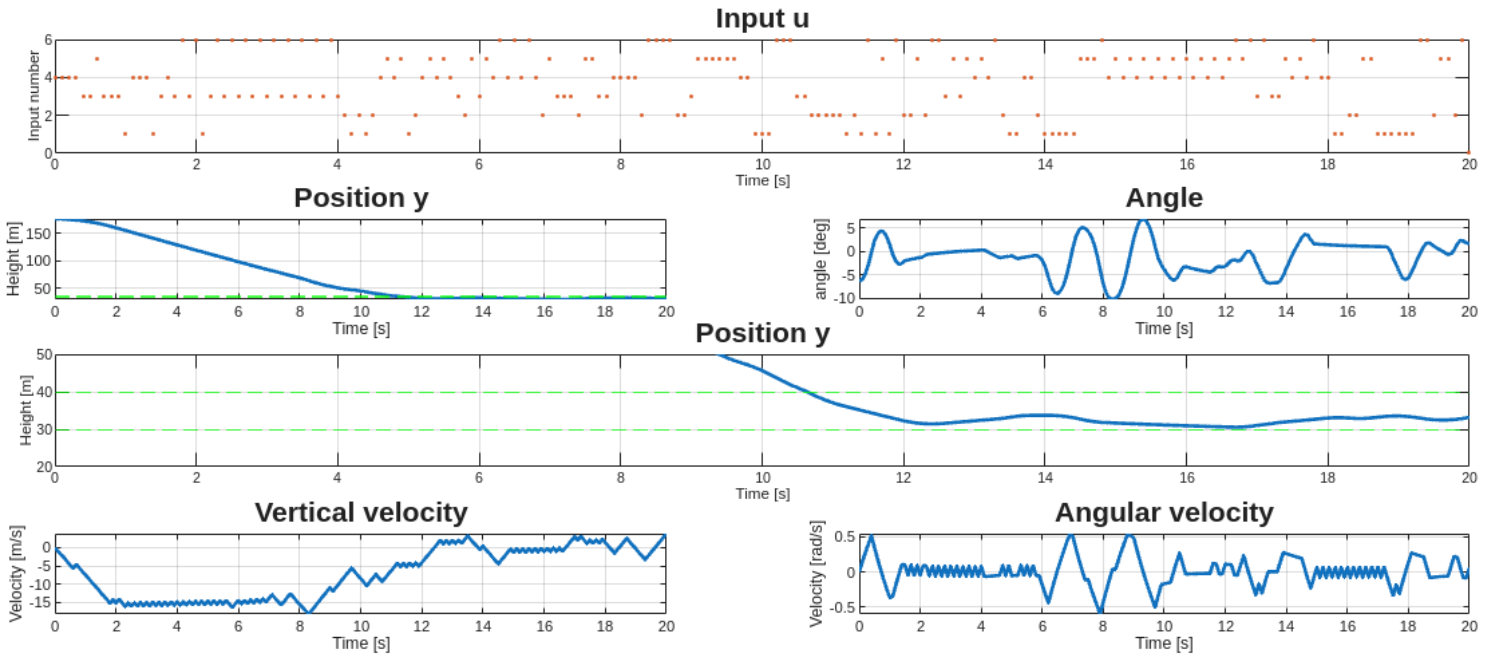
- Position between 150 and 200m
- Angle between -12° and 12°

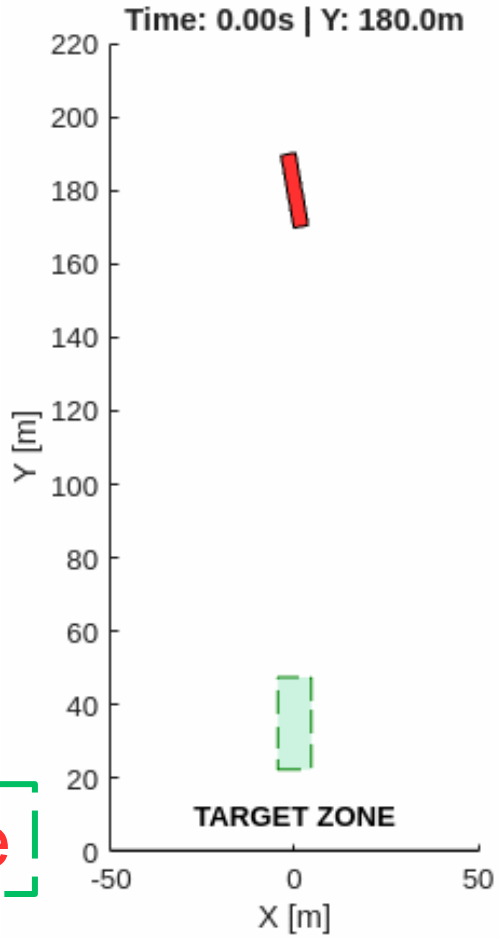
98% of success rate

SARSA - training



SARSA - testing



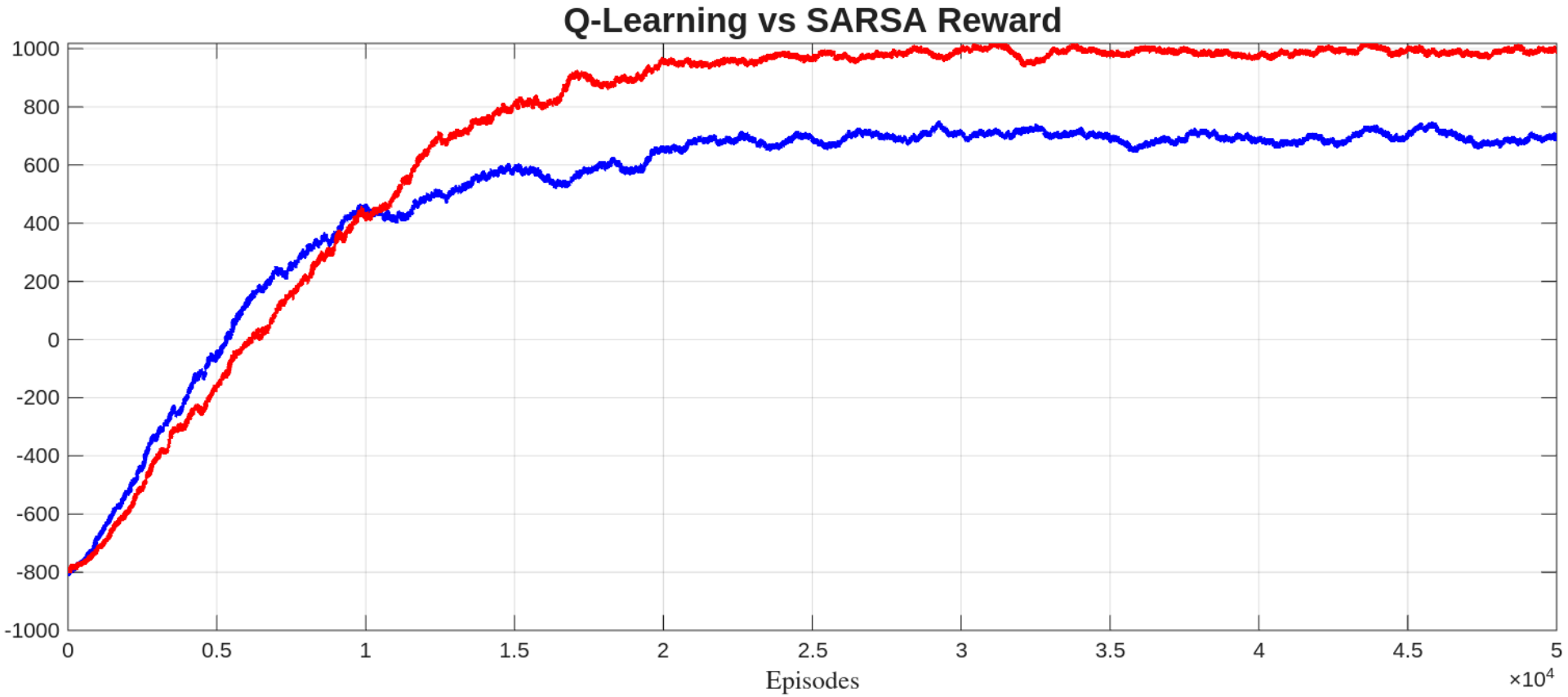


Initial conditions:

- Position between 150 and 200m
- Angle between -12° and 12°

100% of success rate

Q-Learning vs SARSA



Considerations

- The project was challenging to implement with RL: the required discretization is extremely dense (e.g., 12 bins for vertical position)
- Future improvements could include 6D movement (adding the x-axis). However, this expansion is unfeasible with tabular methods and would require function approximation
- The reward function was the key to solve the problem: simple sparse or dense rewards were not enough as the spacecraft often preferred not to land at all

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Thanks for the attention!